

All-in-One Logistics & Financial Statements (FS) System

System Architecture, Logisuite Parity Blueprint, Proof-of-Concept Workflow & 14-Week Master Implementation Schedule (September – December 2026)

CLIENT & SPONSOR	TARGET COMPLETION
Kornet Express Freight Logistics Inc.	December 2026 (Accelerated 14-Week Track)
CORE STRATEGIC GOAL	PROJECT SCOPE
Logisuite Replacement + Seamless FS Integration	Logistics Suite + Financial Statements (Excl. Payroll)
ENGINEERING FOUNDATION	DOCUMENT STATUS
React 18 + Vite / C# .NET Core / SQLite & SQL DB	v1.0 Executive Blueprint & Implementation Plan

Executive Project Alignment

This specification establishes the official engineering workflow and implementation roadmap for Kornet Express. It incorporates a meticulous analysis of all 10 Logisuite operational workflow guides, establishes the integration bridge into our pre-built Financial Statements (FS) engine, and outlines a phased execution schedule targeting full delivery on or before December 2026.

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Kornet Express Logistics → Accounting/FS Integration Flow

Official End-to-End Operational Lifecycle & Automated Financial Posting Architecture

Executive Brief

Key Executive Principle: Operating exactly like Logisuite, operations personnel focus exclusively on booking, dispatching, and cargo manifests. The system **automatically translates** every billable event and operational disbursement into **Financial Statements (FS) General Ledger batches** without manual duplicate re-entry.

Phase 1 • Logistics Operations (Freight & Cargo Intake)

1 Ocean & Air Export/Import

Booking intake, flight/vessel scheduling, container stuffing guides, and MBL/HBL & AWB issuance.

2 P/D Cartage Dispatch

Pick-up/Delivery dispatch orders, barcoded pieces, driver assign, and digital Proof of Delivery (POD).

3 Vehicle Inventory & VIN

1-Click VIN decode (NHTSA API), title clearance validation, and staging warehouse receipts (WR).

4 Dual-Ledger Billing/Cost

Charges captured directly on file: **Billing Tab** (Client freight/THC) & **Cost Tab** (Carrier/port cost).

↓ Shipment Incurred & Margins Audited

Phase 2 • Pre-Audit Margin Gate (File Analysis & Status Lock)

Compares Total Operational Revenue vs. Total Operational Costs. Verifies gross profit $\geq$ threshold (15%). Once validated, file is **CLOSED** to freeze charges before accounting handoff.

SECURITY GATE

↓ Automated Bridge Transfer (Batch Export)

Phase 3 • The Accounting Bridge (Logistics → FS Translation)

A Billing → Accounts Receivable

Generates Official Invoices. Maps freight revenue, fuel surcharges, and doc fees into fs_salebook.

B Disbursements → Accounts Payable

Converts carrier ocean/air line payables, port storage, and cartage costs into fs_purchbook drafts.

C Trial Post Simulation

Automated integrity check verifying **Total Debits == Total Credits** and confirming active fiscal periods before committing.

↓ One-Click Batch Commit to General Ledger

Phase 4 • Financial Statements (FS) System & Live Reports

X General Ledger Posting

Official posting into fs_pournals. Updates real-time debit/credit balances across fs_accounts.

Y Cash Disbursements (CDV)

Automated check voucher printing & remittance generation directly settling carrier/vendor payables.

Z Live Financial Statements

Instant drill-down reporting: **Trial Balance**, **Balance Sheet**, and **Income Statement (P&L)**.

Logistics Operational Event	Integration Bridge Action	FS Accounting Ledger Destination	Financial Statement Impact
Customer Freight Invoiced (Ocean/Air/Cartage)	Auto-Bridge AR	fs_salebook → AR Trade GL (Debit)	Recognizes Operating Revenue on Income Statement & Current Asset on Balance Sheet.
Carrier Freight Incurred (Maersk, Airlines, Drayage)	Auto-Bridge AP	fs_purcbook → AP Trade GL (Credit)	Recognizes Cost of Freight (COGS) & Current Liability on Balance Sheet.
Port & Customs Disbursement (Storage, Duty Advance)	Auto-CDV Voucher	Cash Disbursement Voucher (CDV) Entry	Reduces Cash/Bank account & settles operational vendor liability.
Vehicle Storage / Demurrage Incurred	Accrual Transfer	Direct Journal Voucher (JV)	Accrues staging storage costs against VIN warehouse receipt number.

Kornet Express Freight & Logistics System • Custom Operational & FS Platform Strictly Internal & Executive Review • Confidential

01 Executive Summary & Project Objectives

Kornet Express currently relies on Logisuite for its international freight forwarding (Ocean & Air), domestic cartage (Pickup & Delivery), vehicle inventory, customs clearance, and customer tracking. While Logisuite has historically served the business, its closed legacy architecture limits operational adaptability, multi-user concurrency, real-time web reporting, and mobile cloud access. Kornet Express has commissioned the design and deployment of an **all-in-one in-house Logistics and Financial Statements (FS) web platform** tailored precisely to their operational workflows.

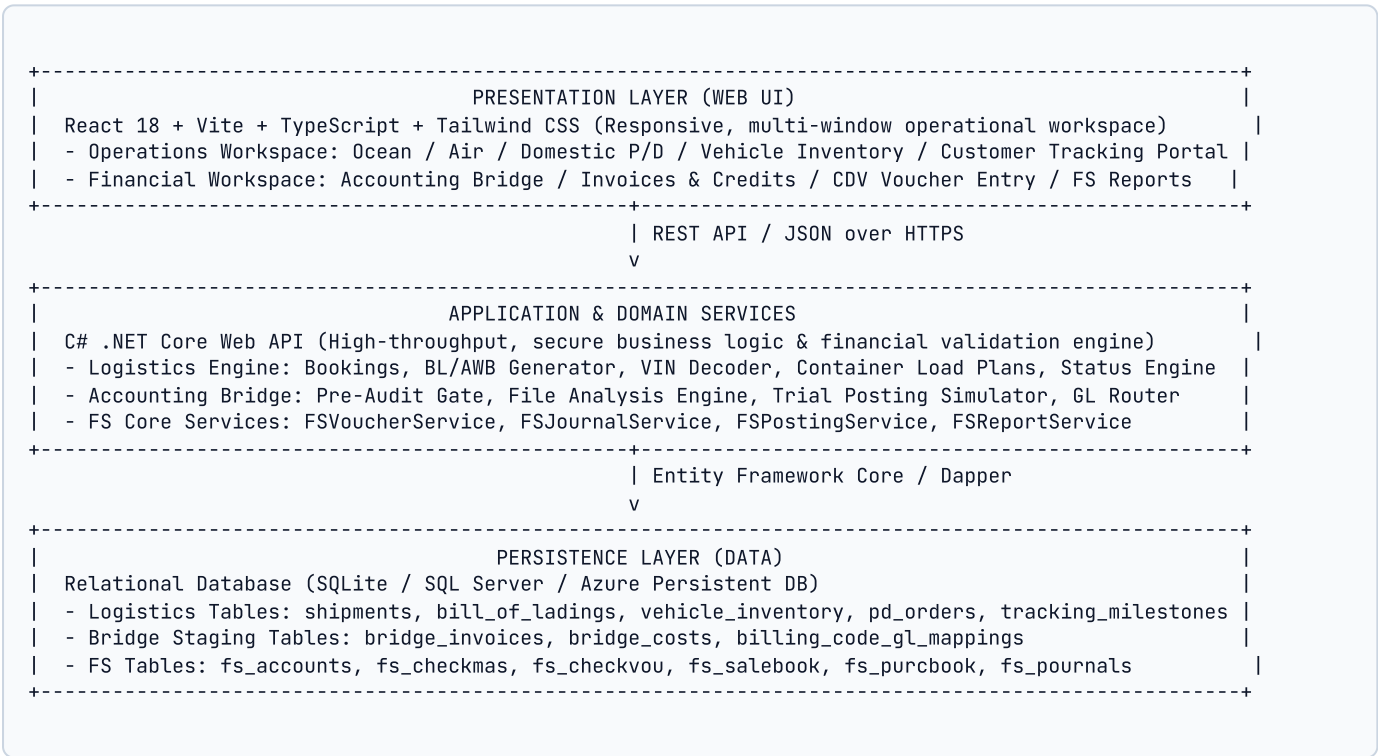
Crucially, rather than building an accounting engine from scratch, this solution directly connects to the **pre-built, production-tested Financial Statements (FS) system** located in our workspace (web-system). This FS core already features complete General Ledger accounts, Cash Disbursement Vouchers (CDVs), Journal Vouchers, Sales/Purchase books, Bank directories, Supplier masters, and statutory financial statements (Balance Sheet, Income Statement, Trial Balance).

Scope Boundary Compliance

In accordance with the client's explicit mandate, the **Payroll module is strictly excluded** from this deployment. The system focuses entirely on **Logistics Operations paired seamlessly with the Financial Statements (FS) engine**.

02 Target Architecture & 3-Tier Enterprise Design

The unified platform is built on a resilient, high-speed 3-Tier Enterprise Web Architecture:



Core Architectural Guarantees

- Dual-Ledger Operational Capture:** Every shipment file concurrently captures customer billing charges (selling rates, prepaid/collect terms) and carrier cost payables (ocean/air freight, chassis fees, drayage, documentation fees).
- Audit-Safe Staging via Accounting Bridge:** Operational personnel never post directly into final financial ledgers. Operational files pass into a staging bridge where accountants audit margins and account codes before committing to GL.

- **Zero Double-Entry:** Billing items flow directly into Sales Journals (fs_salebook) and vendor freight bills flow directly into Purchase Books (fs_purcbook) and Check Vouchers (fs_checkmas).

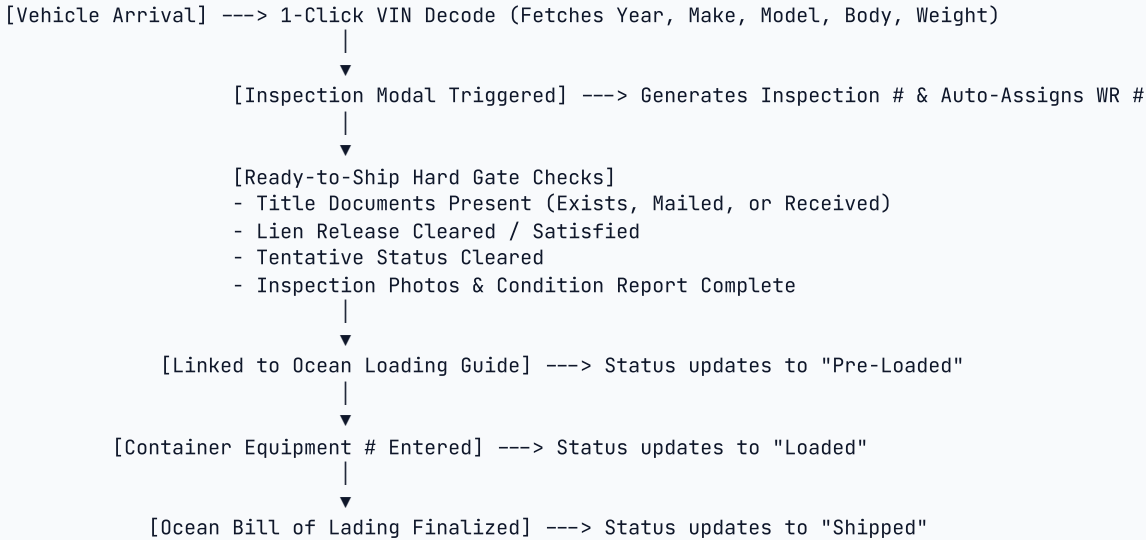
03 Logisuite Operational Parity: Deep Dive into the 10 Modules

Our meticulous analysis of all 10 Logisuite quick guides ensures 100% operational parity for Kornet Express staff:

Module	Operational Workflow & Lifecycle Steps	Deliverables & Accounting Output
Ocean Export 14 Pages Guide	1. Quotes (Selling vs Cost estimates) 2. Bookings (Carrier booking # & internal File # generation) 3. Loading Plan/Guide (Container stuffing & seal #s) 4. B.L. Entry (Master & House BLs, Shipper/Consignee) 5. Manifests (Cargo & Freight) • 6. File Analysis Report 7. Close File • 8. Transfer to Accounting Bridge	Master/House Ocean BOL, Ocean Cargo Manifest, Container Stuffing Sheets • Bridge to AR/AP
Ocean Import 9 Pages Guide	1. File Entry (Direct vs Consolidation, Vessel/Voyage, Ports) 2. Bill of Lading Entry (Charges, Delivery, Broker, Consignee) 3. Close File • 4. File Analysis Report • 5. Transfer to Bridge	Arrival Notices, Delivery Orders, Customs Cargo Release • Bridge to AR/AP
Air Export 9 Pages Guide	1. Quotes • 2. Bookings (Flight & Space reservation) 3. Airwaybill Entry (MAWB/HAWB, Dimensional weight, Rate class) 4. Manifests (Cargo & Freight) • 5. File Analysis Report 6. Close File • 7. Transfer to Accounting Bridge	IATA Standard Airwaybill, Security Manifests • Bridge to AR/AP
Air Import 9 Pages Guide	1. Import Air File (Flight, Origin/Dest, Consignee, Cargo Location) 2. AWB Entry (Prepaid/Collect, Billing charges, Vendor costs) 3. Close File • 4. File Analysis • 5. Transfer to Bridge	Air Cargo Release, Customs In-Bond Notices • Bridge to AR/AP
Pickup & Delivery 12 Pages Guide	1. Order Entry (Xdock, Exchange, Pickup, Delivery, Quote) 2. WR Linking (Cargo search, Vehicle VIN lookup, Barcode scan) 3. Times, Stops, Third Party & Dock Receipt Generation 4. Reports (Labels, Summary, On-Time Performance, Details) 5. Transfer to Accounting Bridge	Barcode Shipping Labels, Driver Run Sheets, Dock Receipts • Bridge to AR/AP
Vehicle Inventory 11 Pages Guide	1. Entry & VIN Decode • 2. Automated Inspection & WR # 3. Title Verification & Lien Checks • 4. Ready-to-Ship Validation 5. Pre-Load to Ocean Guide • 6. Container Loading & Customs Flagging	Vehicle Condition Reports, Dock Receipts, Customs Dossiers • Ocean BL Linkage
Web Tracking 3 Pages Guide	1. Tracking Status Update (Status & Sub-status events) 2. Attachment Upload (POD signatures, Photos, Inspections) 3. Status History Timeline • 4. Tracking Audit Report	Live Public/Customer Tracking Feed & Milestone Alerts
Customer Portal 4 Pages Guide	1. Web Account ID creation tied to Customer Profile 2. Multi-user RBAC (Standard, Agent POD entry, PO Manager)	Self-service client portal for document download & tracking
Invoices & Credits 6 Pages Guide	1. Invoices & Credits Entry (Operational + Misc Invoices) 2. Invoices & Credits Analysis Report (GL & Margin Audit) 3. Trial Post Simulation • 4. Final Post to AR & General Ledger	Commercial Invoices, Credit Memos, AR Ledger entries • FS Integration
Disbursements (AP) 7 Pages Guide	1. Checks Entry (Computer-generated or manual check numbers) 2. Apply to AP Invoices / Discounts / Direct Expense Check 3. Batch Check Printing & Check Voucher Reports 4. Disbursement Audit Report • 5. Trial & Final GL Posting	Physical Check Printouts, CDVs, AP Ledger entries • FS Integration

04 Automotive Logistics & Specialized Kernet Features

Vehicle shipping represents a substantial portion of Kernet Express's freight revenue and involves strict regulatory compliance. The platform replicates and enhances Logisuite's automotive safeguards:



Key Automotive Compliance Mechanisms

- **NHTSA-Powered VIN Decoder:** Instant 17-character VIN decode populating vehicle specifications, minimizing clerical errors.
- **Customs Title Rejection Container Cascading:** When US Customs rejects a title for one vehicle, checking Title Rejected by Customs automatically marks **all other vehicles in the same container**, freezing the container stuffing guide until cleared.
- **Sequential Warehouse Receipt (WR #) Generation:** Completing the physical inspection upon receipt automatically mints the official WR number and issues an automated arrival notice to the customer.
- **Warehouse, Location & Bin Tracking:** Visual bay/bin location management supporting temporary holds, release staging, and warehouse withdrawal logging.

05 The Operations-to-FS Accounting Bridge Integration

The core achievement of this project is linking logistics operations directly to the **already-built Financial Statements (FS) system** without human double-entry:

Phase 1: Pre-Audit Gate

Clerk finalizes shipment. System runs **File Analysis Report** auditing billing vs vendor cost margin. File is locked (**Status: Closed**).

Phase 2: Bridge Staging

Closed file items stage into the Bridge. Customer charges map to AR Sales Journals (fs_salebook). Vendor costs map to AP Purchase Books (fs_purcbook).

Phase 3: Trial & Final Post

Accountant runs **Trial Post** simulation to verify debits/credits. When approved, **Final Post** commits entries into fs_pournals.

End-to-End Operational Ledger Integration

Operational Transaction	Logisuite Bridge Step	Target FS Table ('web-system')	Financial Statement Result
Customer Freight Invoiced (Prepaid/Collect)	Invoices & Credits Transfer	fs_salebook (Sales Journal) • AR Account	Income Statement: Freight Revenue Balance Sheet: Accounts Receivable
Carrier Freight & Handling Costs Accrued	Invoices & Credits Transfer (Cost Tab)	fs_purcbook (Purchase Book) • AP Account	Income Statement: Direct Cost of Sales Balance Sheet: Accrued Accounts Payable
Check / Wire Payment to Carrier / Vendor	Disbursements Posting	fs_checkmas & fs_checkvou (CDV Entry)	Balance Sheet: Cash at Bank (Credit) Balance Sheet: Accounts Payable (Debit)
Customer Payment Collection	Cash Receipts Entry	fs_cashrcpt (Cash Receipts Book)	Balance Sheet: Cash at Bank (Debit) Balance Sheet: Accounts Receivable (Credit)
Batch Posting Execution	Transaction Posting Engine	fs_pournals (Posted GL Master)	Live Trial Balance, General Ledger, Balance Sheet

06 Required FS Engine Adjustments

To ensure our existing FS module functions identically to Logisuite, the following five targeted enhancements will be implemented:

- **Adjustment 1: Accounting Bridge Staging Layer (`AccountingBridgeService.cs`):** An automated queue that receives closed shipments, organizes customer billing lines and vendor cost lines, and presents them in an accountant verification dashboard.
- **Adjustment 2: Operational Cross-Reference Audit Linking:** Enhancing `fs_checkmas`, `fs_salebook`, and `fs_pournals` with `file_no`, `bl_no`, and `container_no` references, enabling 1-click drill-downs from financial reports to operational files.
- **Adjustment 3: Billing Code to GL Account Auto-Router:** An administrative configuration table mapping operational freight codes (e.g. OFRT, FSC, THC, DOCF) to 4-digit FS Account Codes (e.g. 4010 Revenue, 5010 Expense). Staff select logistics codes; the software handles accounting debits/credits.
- **Adjustment 4: Dual-Phase "Trial Post" Simulation Mode:** Replicating Logisuite's [X] Trial Post checkbox in the posting engine. This performs a dry-run balance test, catching unbalanced entries, unmapped codes, or closed fiscal periods before committing changes to `fs_pournals`.
- **Adjustment 5: Pre-Closing File Analysis Report:** An operational audit modal that compares shipment revenue against vendor costs, verifying profit margin before permitting a file to be closed and transferred.

07 Phase 1 Proof-of-Concept (PoC) Prototype Scope

To provide immediate transparency and confirm alignment with Kornet Express, Milestone 1 delivers a fully functional, interactive Proof of Concept (PoC):

1. Shipment & Vehicle Flow

Clerk creates an Ocean/Air shipment and Vehicle Receipt with 1-click VIN decode, customer billing lines, and vendor cost lines.

2. File Analysis & Staging

Clerk runs File Analysis to verify margin, marks file Closed, and transfers charges to the Accounting Bridge.

3. FS Voucher & GL Update

Accountant reviews staged lines, executes Trial Post simulation, performs Final Post, and views live numbers in the Trial Balance.

PoC Interactive Deliverables

- Working React UI integrated directly into the existing web-system shell (Logistics + FS tabs).
- Interactive Vehicle Inventory screen with VIN decoding and Warehouse Receipt generation.
- Ocean/Air shipment creation with dual Billing/Cost ledger tabs.
- File Analysis modal calculating gross profit margin and locking the operational file.
- Automated creation of CDV Check Vouchers (`fs_checkmas`) and Sales Journal lines (`fs_salebook`).
- Real-time Financial Statement update: Trial Balance and Income Statement reflecting the shipment.

08 14-Week Master Implementation Schedule

The project initiates on **September 8, 2026** and concludes with production deployment **on or before December 15, 2026** (a structured 14-week delivery plan):

Phase & Sprint	Timeline	Engineering Deliverables & Activities	Milestone Gate
Phase 1 Sprint 1-3	Sep 08 – Sep 26 Weeks 1-3	- Database schema extensions (logistics tables & bridge queue) - Backend AccountingBridgeService & GL router - Working end-to-end PoC Prototype (Shipment → Bridge → FS Trial Balance)	Milestone 1: Working PoC Demo to Kornet Executives
Phase 2 Sprint 4-6	Sep 29 – Oct 17 Weeks 4-6	- Ocean Export & Import modules (Quotes, Bookings, Stuffing plans, BLs) - Air Export & Import modules (MAWB/HAWB, Dimensional weight, Manifests) - Vehicle Inventory & VIN Decoder engine with WR generation	Milestone 2: Core Freight Operations Functional
Phase 3 Sprint 7-9	Oct 20 – Nov 07 Weeks 7-9	- Domestic Pickup & Delivery (P/D orders, Barcode scan, Dock receipts) - Title rejection container cascading logic & Customs holds - Billing Code to GL Account mapping administrative interface	Milestone 3: Domestic Cartage & Auto Flow Complete
Phase 4 Sprint 10-11	Nov 10 – Nov 21 Weeks 10-11	- Web Status Tracking engine & milestone updates - Customer Web Portal with Role-Based Access (POD download, tracking) - Document printing engine (IATA AWB, Ocean BOL, Barcode labels, CDVs)	Milestone 4: Customer Portal & Print Suite Ready
Phase 5 Sprint 12-14	Nov 24 – Dec 15 Weeks 12-14	- Data migration from legacy Logisuite (Customers, Vendors, Open Files) - End-to-end financial reconciliation & parity testing - User training, administrator manual & production cutover	Milestone 5: Production Go-Live (On or Before Dec 2026)

Weekly Progress Gantt Overview

Weeks 1-3: Architecture & Working PoC Prototype

Weeks 1-3 (PoC)

Weeks 4-6: Ocean & Air Freight Operations + Vehicle Inventory

Weeks 4-6 (Ocean/Air/Vehicles)

Weeks 7-9: P/D Orders, Barcode Scanning & Accounting Bridge Sync

Weeks 7-9 (P/D & Bridge)

Weeks 10-11: Customer Portal, Milestones & Document Generators

Weeks 10-11 (Portal/Docs)

Weeks 12-14: Data Migration, Parity Audits, UAT & Production Launch

Weeks 12-14 (Go-Live & Cutover)

09 Risk Management, Data Security & Migration

Identified Risk	Impact Level	Engineering Mitigation Strategy
Financial Ledger Imbalance	High Impact	Strict database transaction atomicity. The Accounting Bridge rejects any batch where Total Debits do not equal Total Credits.
Post-Transfer Operational Tampering	High Impact	Automated file locking. Once a shipment is closed and bridged, rates, weights, and charges are immutable. Revisions require credit memos.
Legacy Data Migration Quality	Medium Impact	Automated migration validation scripts verifying legacy customer/vendor balances and historical GL figures match exactly to the cent.
User Transition Resistance	Medium Impact	Replicating Logisuite's field navigation patterns, shortcuts, and document printing formats so staff operate with zero productivity loss.

10 Governance, Acceptance Criteria & Next Actions

Formal Acceptance Criteria

- End-to-End Operational Integrity:** Ocean, Air, and P/D shipments successfully generate commercial invoices, carrier payables, and shipping documentation without clerical re-entry.
- Zero Discrepancy Financial Posting:** All bridged transactions post cleanly to fs_salebook, fs_purcbook, fs_checkmas, and fs_pournals, accurately balancing the Trial Balance and Balance Sheet.
- Full Logisuite Parity:** 100% compliance with workflows documented across all 10 guides, including VIN decoding, customs hold cascading, and trial posting.
- On-Time Delivery:** Complete deployment and user training completed on or before December 15, 2026.

Immediate Next Step

Execution of Phase 1 (Proof-of-Concept Prototype): Build and demonstrate the interactive prototype showcasing live shipment booking, File Analysis audit, bridge transfer, and live financial statement update within the existing web-system environment.

Prepared for: Kornet Express Management & Engineering Team

Approved by: Project Steering Committee

Target Go-Live: December 2026